SLI Technology: What Is It?

SLI is an acronym for Scalable Link Interface, a technology originally developed by 3dfx for their Voodoo cards. Now that the company and their cards are history, (after nVidia gulped them out of business) the technology has been improved upon, and a spanking new version is out from nVidia.

In the current scenario, an SLI configuration needs two identical NV45 PCI-Express cards running on a motherboard with two x16 slots. These two cards are connected to each other using a small bridge-type PCB that connects them from above. In this manner, the card connected to the monitor becomes the master and the other card becomes the slave.

The technology aims to provide the user with an increase in graphics performance by rendering an image, using both the GPUs at the same time. The image to be rendered on-screen is automatically load-balanced by the driver, and the image is rendered on the screen. This is done dynamically for each image, and the load on each GPU is accordingly varied. In reviews done by Web sites that had the privilege of laying their hands on an SLI system, the percentage increase is almost 30 to 60 per cent at 1,600 x 1,200 with 4X AA and 8X AF in *Far Cry* depending on the image rendered. That is a difference of more than 20 frames in any given level!

Alienware, a PC major in the US, has developed technology called the Alienware X2 Video Array. The difference here is that the instead of just nVidia video cards, the Alienware array will let you connect any two identical PCI-Express video cards (like ATi X850s). Alienware uses its own proprietary software and a proprietary merger hub that connects both the video cards. This hub synchronises the signals—one scene is collectively rendered by two GPU's. Let's take a look at the cost-to-performance ratio in the present scenario for an SLI rig. The performance increase offered is about 1 to 1.5 times a single card of the same type does. However, the cost increases sharply, since you need to have two top-of-the-line 6800GTs or Ultras to implement this solution—and no, you cannot mix and match! Both cards must be identical in all respects, including memory, and manufactured by the same company.

You will also need to have an efficient cooling system to dissipate the heat that will be generated inside the cabinet by such a setup, not to mention an SMPS of more than 500 Watt to provide for each of the cards drawing power from the SMPS. SLI implementations haven't taken off in a big way because of various issues including buggy drivers. Nevertheless, once these issues are ironed out, SLI may turn out to be the wave of the future!

facturers. The core clock speeds have gone higher, bundled with an increase in the memory speeds from the previous generation and the type of memory used. The biggest addition though, has been the doubling of the pixel pipelines. From 8, it has gone on to 16.

This means that with each clock cycle, the GPU will be able to handle twice the number of pixels and render them on-screen. Given the increased clock and memory frequency of the GPU, a card can chomp through a mind-boggling number of pixels.

The current generation of ATi cards give an increase of more than 50 per cent over the previous generation of cards, in terms of memory bandwidth; while their fill rate (Mtexels) has actually doubled from the previous generation.

SLI has also reappeared on the horizon, and some privileged users actually have a system running SLI. Going by the power that the GPU is being blessed with, it will not come as a big surprise to one day see a video card being compared with the PC processor!

This particular category featured the cream of the current crop. From the X800 to the 6800GT, they all fit into this category. The top-of-the line, most wanted desktop graphic cards on the planet! It's a pity the 6800 Ultra and the Ultra Extreme could not make it to the comparison in time, but the 6800GT and the X800XT PE crowned the high-end category of the comparison.

Their Features

The privilege that high-end cards enjoy is money. For a user who wants to invest in such cards, money is never a concern. Performance is what counts. In addition, from the bundle of software that comes with the cards, it is more than evident that the manufacturers could not agree more.

The games and software bundled with each card is worth at least Rs 1,500, not to mention the extra cables and connectors.

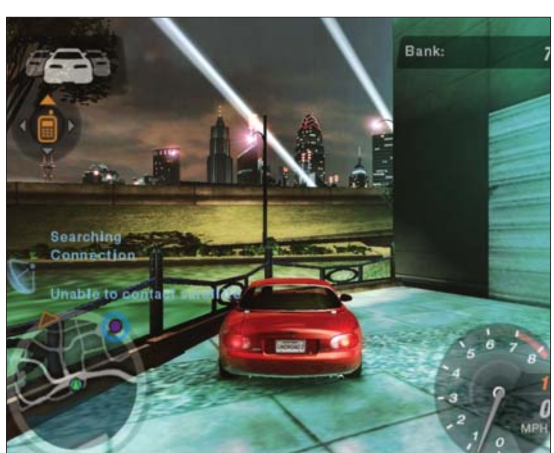
XFX bundled three games with their 6800 GT, while Asus, as usual, bundled three games, a complete CD of game demos, DVD software, drivers, and their proprietary Media Show software. The PowerColor bundled the complete version of *Hitman: Contracts* with their X800 cards, and a multi-utilitarian bag that is a dream for any serious computer user—geek and gamer alike. Gainward was disappointing, and there wasn't much to their package—poor show in the features department.

The rest of the cards also came with commendable packages, but none could match the Asus. Most of the cards had dual DVI connectors on the back panel, for which requisite cables were provided in the package.

The Gainward card looks humongous, with a dual-fan cooling design, and covers up the adjacent PCI slot. This card sucks in air from inside the cabinet and circulates it to efficiently cool the card. But for that, you need a good case with more than enough cooling.

Looking at a RAID setup that most gamers would have, the heat generated inside the system would probably be akin to that of a little furnace.

On the other hand, the XFX 6800 GT is a work of art. Single slot, no fuss, noiseless and sleek-looking. The top of the card is covered by a sleek metal plate,



Need for Speed Underground **on the Gainward GeForce FX 6800GT shines well as per image quality**